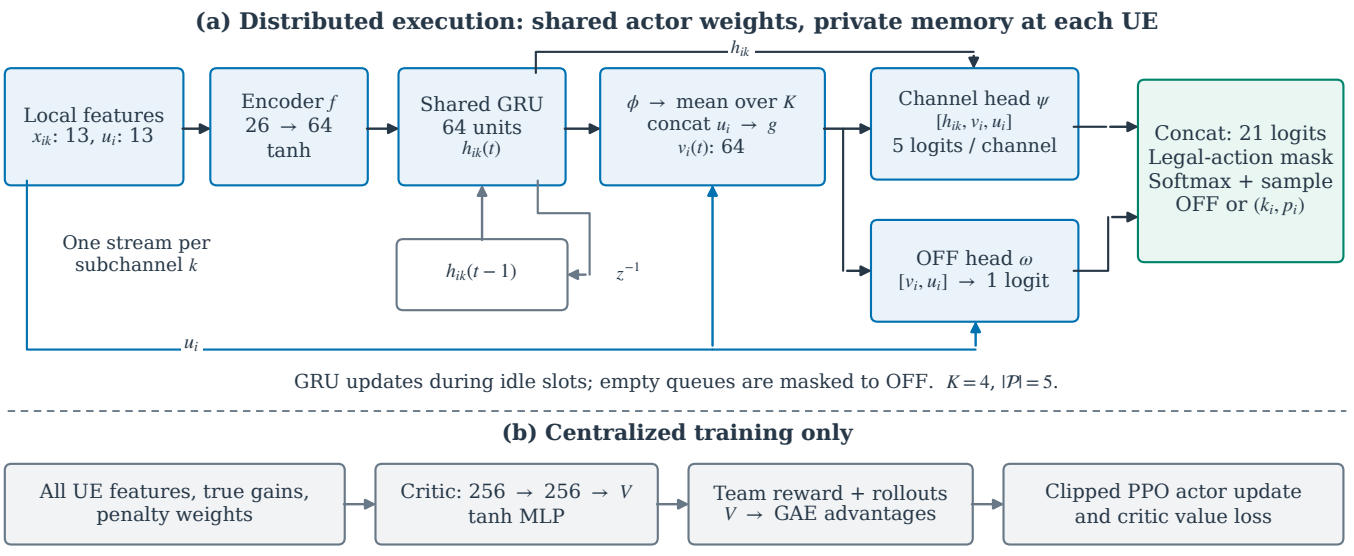
(a) Distributed execution: shared actor weights, private memory at each UE
Local features
x_{ik}: 13, u_i: 13
Encoder f
26 → 64
tanh
Shared GRU
64 units
$h_{ik}(t)$
h_{ik}
ϕ → mean over K
concat u_i → g
$v_i(t)$: 64
Channel head ψ
$[h_{ik}, v_i, u_i]$
5 logits / channel
Concat: 21 logits
Legal-action mask
Softmax + sample
OFF or (k_i, p_i)
One stream per
subchannel k
$h_{ik}(t-1)$
z^{-1}
OFF head ω
$[v_i, u_i]$ → 1 logit
u_i
GRU updates during idle slots; empty queues are masked to OFF. $K = 4$, $|\mathcal{P}| = 5$.
(b) Centralized training only
All UE features, true gains,
penalty weights
Critic: 256 → 256 → V
tanh MLP
Team reward + rollouts
V → GAE advantages
Clipped PPO actor update
and critic value loss